



UDARA SHANUKA

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ABOUT MYSELF

Focused on building scalable backend systems and DevOps-driven workflows. Experienced in developing and deploying applications with a strong focus on performance, automation, and maintainability. Comfortable working across software and infrastructure environments, including CI/CD, cloud platforms, and containerized deployments. Strong team player who enjoys collaborative development and contributes effectively to group environments. Passionate about problem-solving with a practical, self-driven approach and continuous learning mindset.

EDUCATION

Bachelor of Science (Hons) in Electronics and Computer Science - University of Kelaniya, Sri Lanka | 2022 – Present

Related Coursework: Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Database Management Systems, Data Communications and Networks, Web & Internet Technologies, Cyber Security & Forensics, Full-Stack Development, Machine Learning, Intelligence Systems.

TECHNICAL SKILLS

- **DevOps & Cloud:** Docker, Kubernetes, Helm, ArgoCD, GitHub Actions, Jenkins, Terraform, Ansible, AWS, GCP, Azure, Linux
- **Backend:** Node.js, Express.js, Spring Boot, REST APIs, Prisma ORM, Go (learning), Axios
- **Frontend:** Next.js, React, TypeScript, JavaScript, Angular, Zustand, TanStack Query
- **Programming Languages:** Java, JavaScript, TypeScript, Python, C
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Tools:** Git, Postman, VS Code, IntelliJ, Jest, ESLint, Husky

PROJECTS

University Multi-Portal Management System - SSO + DevOps Deployment ([Git](#)) Personal | 2025 (Dec) - 2026 (Jan)

Next.js, Turborepo, TypeScript, Prisma, PostgreSQL, K3s, Terraform, ArgoCD, Ansible, WSO2 Identity Server

- Built a multi-portal system (Student, Library, Admin) with a shared component architecture using Next.js and Turborepo.
- Implemented SSO authentication and role-based access control using WSO2 Identity Server.
- Designed scalable backend services with TypeScript, Prisma, and PostgreSQL for multi-tenant data management.
- Automated Azure infrastructure provisioning using Terraform and deployed workloads on a K3s cluster.
- Established GitOps CI/CD pipeline using ArgoCD and Ansible for automated, zero-downtime deployments.
- Integrated monitoring stack with Prometheus and Grafana for system observability.

TrekDesk AI - AI-Powered Tour Booking Assistant (Backend + Realtime System) ([Live site](#)) Personal | 2026 (March)

React, Node.js, Express, TypeScript, WebSockets, PostgreSQL, pgvector, Gemini Live API, Google APIs

- Built a backend system for an AI-driven tour booking assistant supporting real-time chat and voice interactions.
- Developed a RAG pipeline using PostgreSQL + pgvector for semantic search over tour and booking data.
- Integrated Gemini Live API with WebSockets for low-latency conversational AI (text + voice).
- Implemented AI tool-calling for dynamic actions like availability checks and booking workflows.
- Integrated Google Calendar API for automated scheduling and guide availability management.
- Built secure Google OAuth authentication with JWT-based session handling.
- Designed embeddable widget system with domain validation for third-party website integration.

Trekking Tours - Commercial Web Application ([Live Site](#))

Personal | 2026 (Feb)

Next.js, Vercel, Resend, Analytics

- Developed and deployed a production website with real customer inquiries.
- Integrated analytics to analyze visitor behavior and improve engagement.
- Delivered SEO-optimized, responsive UI for enhanced reach.

Cloud-Native API Platform with WSO2 API Management([Git](#))

Personal | 2025 (Nov)

Node.js, Kubernetes, Docker, Helm, WSO2 API Manager, OpenAPI 3.0, MySQL

- Built a cloud-native backend API system for ecommerce domain services (catalog, cart, orders).
- Implemented API-first design using OpenAPI 3.0 with WSO2 API publishing and documentation.
- Integrated WSO2 API Gateway for OAuth2/JWT authentication, rate limiting, and policy enforcement.
- Deployed containerized services on Kubernetes using Helm for scalable and reproducible environments.
- Designed modular backend architecture and persistent database storage (PVCs).

RosaLover - E-Commerce Platform ([Git](#))

Group | 2024 (Oct) – 2025 (Janu)

React, Spring Boot, MySQL, Stripe

- Designed and developed an e-commerce website with features like product management, wishlist, cart, and payment gateway.
- Built an admin panel for managing inventory, orders, and user roles.
- Contributed to frontend and backend design and system integration.

AI-Driven 3D Posture Detection in Olympic Weightlifting (Research Project) ([Git](#))

2024 (Jun) – 2025(June)

Python, TensorFlow, OpenCV, MediaPipe, LSTM, Automation Pipelines

- Designed an automated pipeline for multi-camera video processing, including frame extraction, pose estimation, joint angle computation.
- Implemented Python-based feature engineering pipelines to automate dataset creation from raw videos.
- Developed CNN–LSTM models for phase classification and real-time feedback generation in Snatch & Clean-and-Jerk.

Movie Booking Systems (Java & C) [Git](#)([Java](#), [C](#))

Group | 2022 (Aug) – 2022(Oct) | 2023(Feb) -2023(Mar)

Java (Swing GUI, OOP), C (Functional Programming)

- Developed a movie booking system in **Java** displaying OOP principles with GUI for user interaction.
- Built a separate **C implementation** focusing on procedural programming and data structures.

CERTIFICATIONS

- **Front End Development Libraries** – FreeCodeCamp | Aug 2024 |
- **Server-Side Web Programming** – University of Moratuwa | Jul 2024 | Credential ID: TYHBqa2SrZ
- **Front End Web Development** – University of Moratuwa | May 2024 | Credential ID: 0mzOfqedS4
- **Web Design for Beginners** – University of Moratuwa | Jan 2024 | Credential ID: cv8CQX5gBC
- **JavaScript Algorithms and Data Structures** – FreeCodeCamp | Dec 2023
- **Responsive Web Design** – FreeCodeCamp | Aug 2023

REFERENCE

Professor A.L.A.K. Ranaweera

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